

1 Introduction

This project studies portfolio-level clustering for U.S. small-cap equities. We compare three complementary approaches—(i) Path-Signature features with K-means, (ii) correlation-based hierarchical clustering, and (iii) Dynamic Time Warping (DTW) on cumulative log returns—to uncover structure in small-cap return dynamics. Unlike single-name idiosyncrasies or one-shot cross-sectional snapshots, our focus is on rolling time windows, allowing us to track how clusters form, split, and migrate across regimes (e.g., risk-on vs. risk-off, liquidity squeezes, rate shocks). The end goal is to deliver clusters that are both statistically robust and financially interpretable—for instance, distinguishing quality vs. junk, commodity-linked names, or growth/long-duration profiles inside the small-cap universe.

2 Motivation

Small caps are an attractive but messy corner of the market. They exhibit higher volatility, weaker analyst coverage, thinner liquidity, and sharper microstructure noise than large caps. Classical correlation clustering often collapses these effects into one factor (the “market mode”), missing path-shape information such as “fast crash then rebound,” volatility timing, and asymmetric drawdowns that practitioners care about. Path-Signature features explicitly encode the trajectory of returns (and auxiliary channels like realized volatility), offering a richer representation than correlation alone. DTW adds a shape-alignment baseline to test whether timing differences—not just co-movement—drive the grouping.

From a portfolio-construction standpoint, small-cap clustering can:

- Improve diversification by avoiding look-alike exposures that standard sector tags fail to capture.
- Support regime-aware rebalancing, as clusters reconfigure under liquidity stress or rate cycles.
- Enhance risk diagnostics, surfacing latent groups (e.g., high-beta and low-profitability “junk” pockets) that inflate tail risk.

In short, we are motivated to test whether Signature-based clustering delivers more stable, interpretable, and actionable groupings in small caps than traditional correlation or DTW methods—especially when evaluated through rolling windows, stability metrics, and simple economic diagnostics (e.g., factor loadings, drawdown profiles).

3 Data

Our analysis combines three datasets: CRSP security-level equities data, the Ken French Data Library’s 49 Industry Portfolios (Daily), and an equity news sentiment feed from Alexandria Technology.

The core panel is drawn from the Center for Research in Security Prices (CRSP). We use CRSP’s security identifiers (PERMNO/PERMCO), ticker, company name, exchange code (EXCHCD), security code (SHRCD), four-digit SIC industry code (SICCD), price (PRC), shares outstanding (SHROUT), and daily simple return (RET). This panel allows us to define a point-in-time universe of U.S. common stocks (NYSE/AMEX/NASDAQ) in a target market-capitalization range of 2–10 billion USD at designated anchor dates, then observe those same securities at daily frequency within the study windows. CRSP is accessed through Wharton Research Data Services (WRDS); replication simply requires querying the same fields for the same anchor dates and windows and recording the WRDS pull date.

To contextualize and interpret clustering outcomes, we also rely on the Ken French Data Library’s 49 Industry Portfolios (Daily) file. This dataset reports daily total returns for forty-nine value-weighted U.S. industry portfolios, where industries are defined from four-digit SIC codes and membership is reassessed each June for the subsequent July–June period. The file is organized with trading dates in YYYYMMDD format and one column per industry. It is publicly available from the Ken French Data Library website, which also provides the accompanying “Details” documentation and industry-definition materials. Citing the download date enables exact replication from the same snapshot.

Finally, for news-based annotations we use Alexandria Technology’s Equity Newswires feed. The vendor delivers time-stamped, security-linked news records with fields such as headline/source identifiers, ticker and ISIN, directional sentiment scores with model confidence, novelty/relevance flags, and event tags. This feed is obtained directly from Alexandria under license; replicators should request the same product and historical window and note the vendor delivery date and schema version.

Throughout the paper we report the data sources, file versions (or query dates), and the calendar windows used, so that an independent researcher with access to CRSP/WRDS, the Ken French website, and the Alexandria feed can retrieve the same snapshots and reconstruct the panels used in our experiments.

4 Methodology and Process

4.1 Ledoit–Wolf Correlation-Based Clustering

The first clustering approach relies on the linear dependence structure of asset returns. Let $R \in \mathbb{R}^{T \times N}$ denote the matrix of daily returns, where T is the number of time periods and N is the number of assets. To obtain a stable estimate of the covariance matrix in the presence of limited observations and high dimensionality, we use the Ledoit–Wolf shrinkage estimator as implemented in the `scikit-learn` library. This estimator automatically determines an optimal shrinkage intensity and returns a regularized covariance matrix $\hat{\Sigma}_{LW}$, which reduces estimation noise while retaining essential correlation structure.

The corresponding correlation matrix is computed as

$$\rho_{ij} = \frac{\hat{\Sigma}_{LW,ij}}{\sqrt{\hat{\Sigma}_{LW,ii}\hat{\Sigma}_{LW,jj}}},$$

where ρ_{ij} measures the degree of linear co-movement between assets i and j . The correlation matrix is then transformed into a distance matrix

$$D_{ij} = \sqrt{2(1 - \rho_{ij})},$$

so that more highly correlated assets are closer in distance space. Hierarchical agglomerative clustering with Ward linkage is then applied to D to form groups of assets that exhibit similar correlation structures. This method captures relationships driven primarily by linear dependencies in return dynamics and serves as a baseline for comparison with nonlinear approaches.

4.2 Signature-Based Clustering

The second clustering framework leverages the rough path *signature* representation to capture nonlinear and temporal characteristics of asset return trajectories. Unlike correlation-based methods, which summarize linear co-movements, this approach embeds the entire path evolution of each asset into a high-dimensional feature space derived from iterated integrals.

Each asset's cumulative return path is first constructed as

$$P_t^i = \prod_{\tau \leq t} (1 + r_\tau^i),$$

and normalized to start at one to ensure comparability across assets. To preserve the temporal ordering of price movements, we apply a *lead-lag transformation*, which maps each one-dimensional time series into a two-dimensional path that encodes both the current and lagged values. This transformation enhances the representation of sequential structure, making it suitable for computing path signatures.

We then compute the truncated path signature of each transformed trajectory using the `iisignature` library. For a continuous path $X : [0, T] \rightarrow \mathbb{R}^d$, the signature of order m is defined as the sequence of all iterated integrals:

$$S^{(m)}(X) = \left(1, \int dX_t, \int dX_{t_1} \otimes dX_{t_2}, \dots, \int_{0 < t_1 < \dots < t_m < T} dX_{t_1} \otimes \dots \otimes dX_{t_m} \right).$$

In our implementation, we truncate the expansion at depth $m = 4$, balancing representational richness and computational tractability. Each signature is then flattened into a feature vector, forming a feature matrix

$$\mathbf{S} = [S^{(4)}(X^1), S^{(4)}(X^2), \dots, S^{(4)}(X^N)]^\top,$$

where each row corresponds to one asset.

We perform agglomerative hierarchical clustering with Ward linkage directly on the Euclidean distances in this signature feature space. This groups assets whose return trajectories exhibit similar temporal shapes and nonlinear behaviors, even if their linear correlations differ. Visualization through t-SNE projection of the signature features provides geometric insight into the structure of these clusters, while cumulative return analysis reveals how each cluster evolves over time.

5 Experiments

5.1 First Experiment: Optimal Cluster Selection based on Silhouette Score

We conducted an initial experiment to compare the Ledoit–Wolf correlation-based clustering and the signature-based clustering methods using ten years of daily return data from 2015 to 2025. For both approaches, the optimal number of clusters was determined using the silhouette coefficient, which evaluates clustering performance by balancing intra-cluster cohesion and inter-cluster separation.

The silhouette analysis indicated that both methods achieved their highest scores when the number of clusters was set to two. This suggests that, under both linear correlation and path-signature representations, the asset universe naturally partitions into two dominant regimes. However, the structural characteristics of the clusters produced by each method differ substantially.

The Ledoit–Wolf correlation-based clustering formed two stable and economically interpretable groups, reflecting similarity in linear co-movement and persistent correlation patterns among assets. By contrast, the signature-based clustering—though designed to capture nonlinear and temporal dependencies—did not yield an ideal partition in this initial experiment. The clusters derived from the signature features appeared less distinct and exhibited weaker separation in terms of return trajectories and sectoral alignment.

This result implies that, while rough path signatures provide a richer representation of time series dynamics, their direct use with Euclidean distance in hierarchical clustering may not fully exploit the expressive capacity of the signature space.

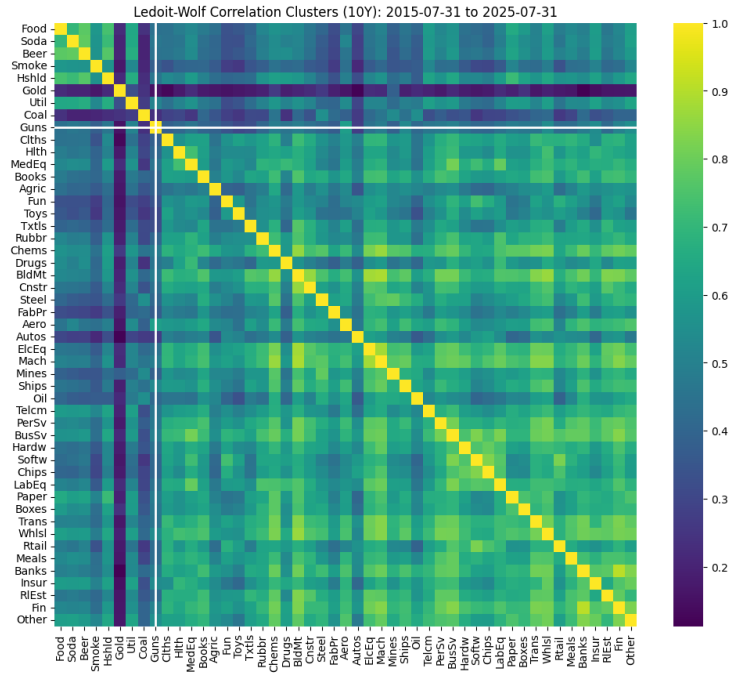
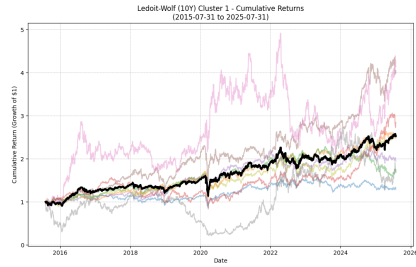
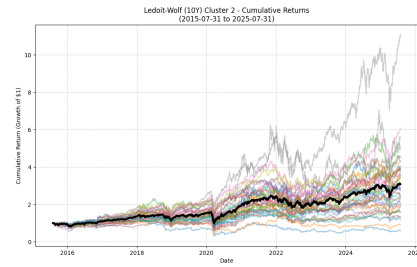


Figure 1: Ledoit–Wolf correlation matrix between industry portfolios.



(a) Cumulative returns of assets within the first cluster and their average.



(b) Cumulative returns of assets within the second cluster and their average.

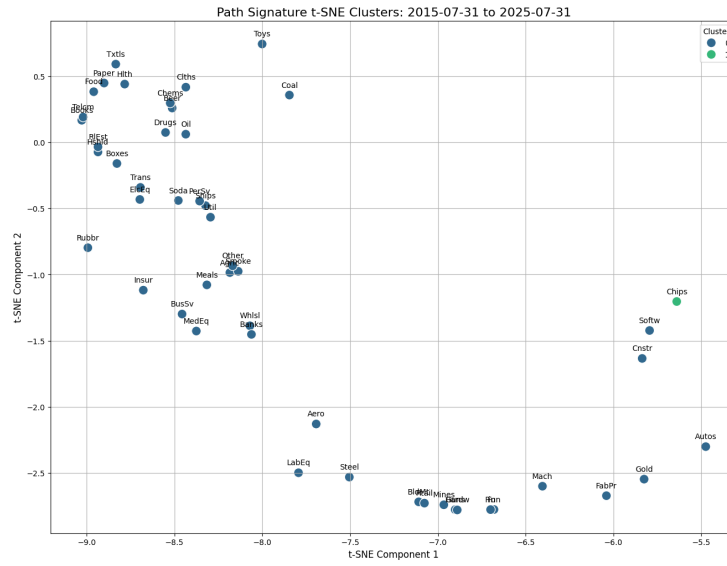
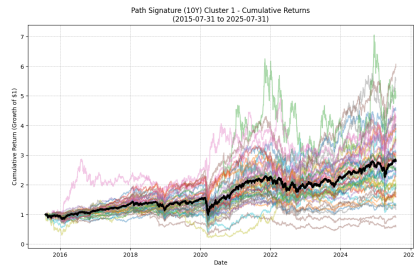
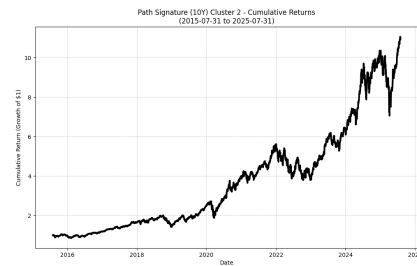


Figure 2: Signature-based industry portfolio clusters.



(a) Cumulative returns of assets within the first cluster and their average.



(b) Cumulative returns of assets within the second cluster and their average.